

D 120207

(Pages : 2)

Name.....

Reg. No.....

**SIXTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
MARCH 2025**

Physics/Applied Physics

PHY6B13/APH6B13—RELATIVISTIC MECHANICS AND ASTROPHYSICS

(Admissions Year—2019 Onwards)

Time : Two Hours

Maximum : 60 Marks

*The symbols used in this question paper have their usual meanings.***Section A (Short Answer Type)***Answer **all** questions in two **or** three sentences, each correct answer carries a maximum of 2 marks.*

1. What are the significances of negative results in Michelson-Morley experiment ?
2. Distinguish between space like and time like intervals.
3. Explain the significances of special theory of relativity.
4. State the principle of equivalence.
5. What is Chandrasekhar limit ?
6. Establish the relation between apparent magnitude and brightness ratio.
7. Explain the method 'stellar parallax'.
8. What are pulsating stars ?
9. Give a note on proton-proton chain reaction.
10. Explain the mechanism of star formation triggers.
11. State Hubble's law.
12. Two stars, α -Canis Majoris and o-Ceti, have a temperature of 9200 K and 1900 K, respectively. What are their peak wavelengths ?

(Ceiling = 20 marks)

Turn over

Section B (Paragraph / Problem Type)

Answer **all** questions in a paragraph of about **half a page** to one page, each correct answer carries a maximum of 5 marks.

13. Obtain the Lorentz transformation equations.
14. Explain the doppler shift in sound.
15. Explain the big bang cosmology.
16. Outline the significances of H-R diagram.
17. Explain the features of Planetary Nebulae.
18. Discuss the features of gravitational lensing.
19. Explain the evolution of white dwarfs.

(Ceiling = 30 marks)

Section C (Essay Type)

Essays - Answer in about **two pages**, any **one** question.

The question carries 10 marks.

20. Derive the relativistic mass-energy relation.
21. Explain the terms :
 - (1) Neutron stars ; (2) Black holes ; and (3) Pulsars.

(1 × 10 = 10 mark)